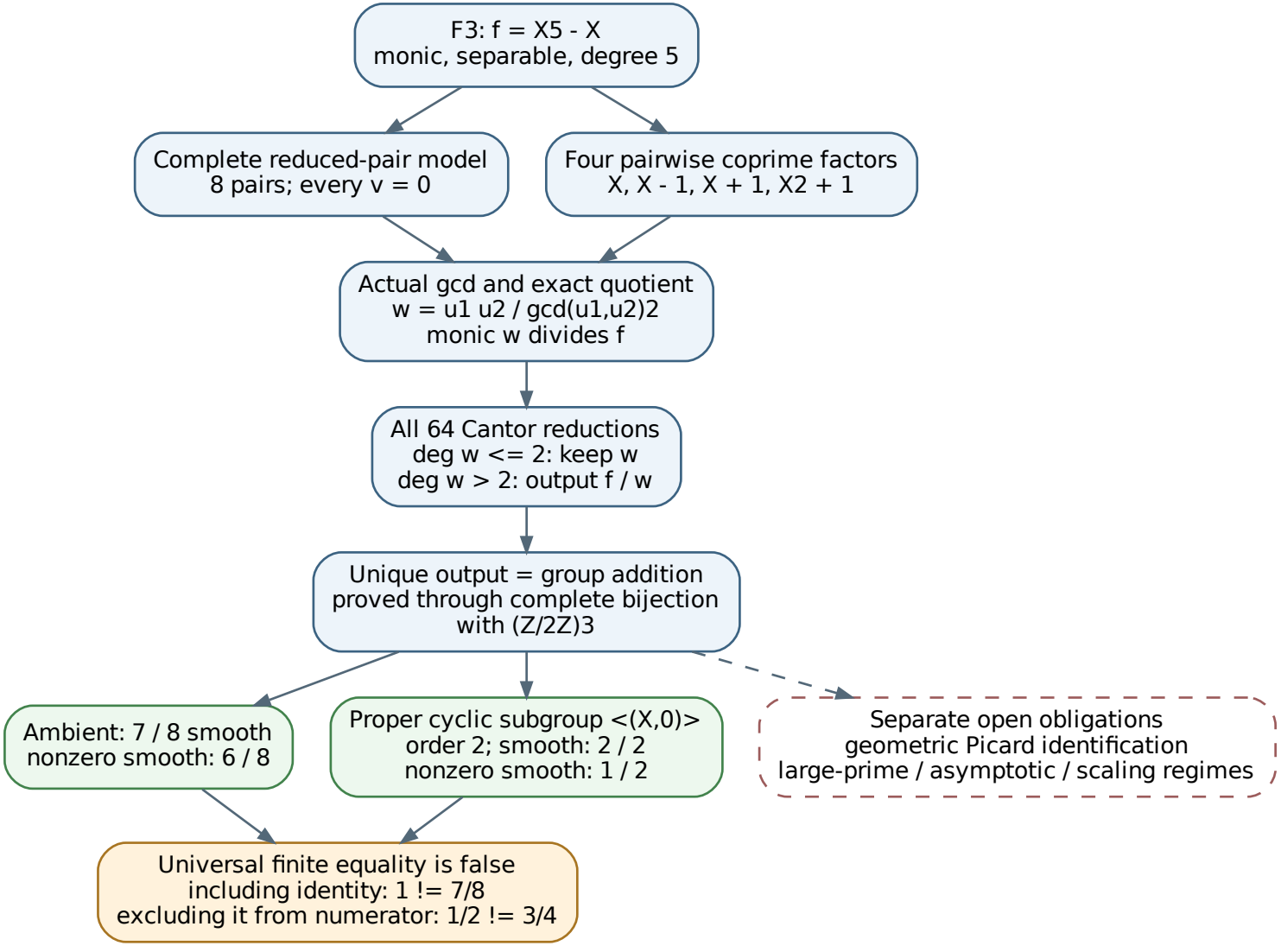
F3: f = X5 - X
monic, separable, degree 5
Complete reduced-pair model
8 pairs; every v = 0
Four pairwise coprime factors
X, X - 1, X + 1, X2 + 1
Actual gcd and exact quotient
w = u1 u2 / gcd(u1,u2)2
monic w divides f
All 64 Cantor reductions
deg w <= 2: keep w
deg w > 2: output f / w
Unique output = group addition
proved through complete bijection
with (Z/2Z)3
Ambient: 7 / 8 smooth
nonzero smooth: 6 / 8
Proper cyclic subgroup <(X,0)>
order 2; smooth: 2 / 2
nonzero smooth: 1 / 2
Separate open obligations
geometric Picard identification
large-prime / asymptotic / scaling regimes
Universal finite equality is false
including identity: 1 != 7/8
excluding it from numerator: 1/2 != 3/4